

Rajat Thakur

Senior Software Engineer | Voice AI & Conversational Platforms | Real-Time Telephony

Noida, India • thakurrajat126@gmail.com • [linkedin.com/in/thakurrajat126](https://www.linkedin.com/in/thakurrajat126)

PROFESSIONAL SUMMARY

Backend and Voice AI engineer with 8+ years building real-time telephony and conversational-AI systems at production scale. Specializes in low-latency LLM voice agents over Asterisk/PJSIP and Kamailio SIP infrastructure, with deep experience in Indic-language STT/TTS pipelines and RAG. Has shipped production voice bots across banking, and enterprise deployments, backed by strong fundamentals in Java/Spring Boot, Python, and resilient distributed systems.

CORE SKILLS

Languages: Java, Python, SQL

Voice & Telephony: Asterisk / PJSIP, Kamailio, ARI, SIP / RTP / SRTP, SMPP, AudioSocket, SIPp, SIP trunking

Voice & Conversational AI: Pipecat, Dify, STT/TTS, VAD, multi-agent systems, MCP, LangChain

RAG & Data Stores: Redis / Redis Stack (RAG), MySQL, MongoDB

Messaging & Infra: ActiveMQ, Nginx, Docker

Cloud & Observability: AWS (EC2/GPU, IAM, Bedrock), GCP (Cloud Run), Grafana, ELK / Kibana

PROFESSIONAL EXPERIENCE

Senior Software Engineer — Sinch

Dec 2021 – Present

Noida, India · Voice AI platform for Indian SMB & enterprise CPaaS clients

- **Built a real-time Indic-language Voice AI platform** integrating Asterisk/PJSIP + Pipecat + AudioSocket with STT/TTS, supporting conversational voice bots in Hindi, Telugu, Tamil and other regional languages.
- **Architected centralized SIP infrastructure on Kamailio** for trunk registration and routing with media handling and multi-Asterisk HA, sustaining ~1300–1800 concurrent calls per 16-core node and validated via SIPp load testing.
- **Cut voice-agent response latency** by deploying GPU-served inference (NVIDIA T4 / g4dn) and by diagnosing and resolving time-to-first-token (TTFT) and time-to-first-audio (TTFA) bottlenecks on AWS.
- **Built Indic-language RAG pipelines** on Redis-based vector search with multilingual embeddings.
- **Designed a Vendor Routing Manager** that automated multi-vendor failover, improving call-routing availability and resilience for enterprise-scale deployments.
- **Earlier scope:** developed Voice Services middleware (IVR, missed-call, click-to-call) and integrated Asterisk IVR via ARI APIs for automated, AI-driven call handling across the CPaaS domain.

Senior Software Engineer — OneBill

Sep 2017 – Dec 2021

Bengaluru, India · UIDAI national-scale messaging & billing systems

- **Built and deployed a high-throughput SMS Gateway middleware** for the UIDAI program, handling national-scale transaction routing, real-time monitoring, and billing across multiple data centers using Java, SMPP, ActiveMQ, Nginx, and MongoDB.
- **Engineered multi-datacenter Disaster Recovery** with real-time traffic distribution and near-zero loss, ensuring mission-critical reliability for a national-scale service.
- **Established production benchmarking and observability** with Grafana and ELK (Kibana), improving system visibility and enabling targeted performance tuning.
- **Enhanced a flexible Workflow Engine** with third-party integrations and SSO between billing and workflow systems, and shipped user-centric UI/backend features that streamlined operations.

EDUCATION

B.Tech, Computer Science — Baddi University of Emerging Science and Technology (2012 – 2016)